Geared Lolly Dispenser Teacher concept guide

Year 9 Engineering | Revised 30 October 2026

Use this guide with the revised 50-page student workbook and the current four website toolkits. Model reasoning and expected evidence are provided for the written, drawing, design, build and evaluation tasks. Accept equivalent accurate explanations in students' own words.

No calculations are required. Class plans, dimensions, materials, machine processes, safety checks, food-contact suitability and formal assessment conditions remain teacher-controlled. Open records must contain genuine student observations; the examples below do not assert that a trial, approval or improvement occurred.

Print the student workbook as 50 A4 pages, or 25 double-sided sheets on the long edge. The PDF retains the current published cover; the native editable Word retains its existing cover composition. Both contain the same revised 49 interior pages.

Student page 2

Write one question about how a lolly dispenser works.

Accept a relevant question about the mechanism, such as how the handle moves the release part, why gears turn in different directions, or what causes a jam. The student's own question is valid; no fixed wording is required.

Activity table: Learning route / Pages / Main evidence

Read-only learning-route reference. Check that the student can locate the relevant mechanism/design/manufacture/testing pages; no written answer required.

Activity table: Drawing or response space

Accept a relevant question about the mechanism, such as how the handle moves the release part, why gears turn in different directions, or what causes a jam. The student's own question is valid; no fixed wording is required.

Student page 3

Record the instructions your teacher has confirmed, or a question you need to ask.

Record only the actual teacher-confirmed task, submission route and project scope. If a fact is not confirmed, record the question to ask. A blank or honest unresolved question must not be replaced with an invented instruction.

Name one thing your dispenser must do and one limit you need to work within. How will you check them?

Name one requirement from the actual brief and one genuine constraint. A requirement might concern the intended release or safe operation; a constraint might be supplied material, available space or parts. Explain a relevant check. Specific dimensions come from the teacher's plan.

Activity table: Item / Teacher-confirmed instruction or question

Record only the actual teacher-confirmed task, submission route and project scope. If a fact is not confirmed, record the question to ask. A blank or honest unresolved question must not be replaced with an invented instruction.

Activity table: Drawing or response space

Name one requirement from the actual brief and one genuine constraint. A requirement might concern the intended release or safe operation; a constraint might be supplied material, available space or parts. Explain a relevant check. Specific dimensions come from the teacher's plan.

Student page 4

Why might a dispenser turn smoothly but still release too many lollies?

Turning movement alone does not separate lollies. The opening, gate or feed path may let more than the intended amount through.

Activity table: Drawing or response space

Turning movement alone does not separate lollies. The opening, gate or feed path may let more than the intended amount through.

Student page 5

A small driver turns a larger driven gear. Describe the output as faster or slower, and name its direction compared with the driver.

The larger driven gear turns more slowly and, for an external pair, in the opposite direction.

Activity table: Smooth movement matters Suitable gears need the correct positions and fit. Use the approved plan. If a handle becomes stiff, stop and ask for help rather than forcing it.

Read-only concept callout. Expected understanding: gears need suitable fit, and a stiff handle calls for stopping and teacher help rather than force. No pitch-circle calculation required.

Activity table: Drawing or response space

The larger driven gear turns more slowly and, for an external pair, in the opposite direction.

Student page 6

What would you look for if the handle suddenly became difficult to turn?

Look for rubbing, gears too tight, a crooked shaft or a displaced part. Stop and ask the teacher before adjusting it.

Activity table: Check the fit Gears need to mesh without rubbing or slipping. Check where shafts are supported and whether parts are held in position. Use the teacher's spacing and fit instructions; do not guess an allowance.

Read-only fit callout. Expected understanding: suitable tooth engagement and supported shafts allow smooth movement. Actual spacing comes from the plan.

Activity table: Drawing or response space

Look for rubbing, gears too tight, a crooked shaft or a displaced part. Stop and ask the teacher before adjusting it.

Student page 7

How can you tell the difference between a simple idler train and a compound train?

An idler train has an extra separate gear; a compound train has two gears fixed to the same shaft.

Activity table: Drawing or response space

An idler train has an extra separate gear; a compound train has two gears fixed to the same shaft.

Student page 8

Which arrangement is easiest for you to explain? Sketch it and label the driver, driven gear and shafts.

Accept an accurate labelled sketch of any described arrangement with a clear reason; no ratio required.

Activity table: Arrangement / What connects? / What to notice

Read-only arrangement comparison. One external pair reverses direction; a simple three-gear line makes first and last turn in the same direction; compound trains are identified by gears sharing a shaft. No arithmetic required.

Activity table: Look at the shafts An idler is an extra gear between driver and driven gears. A compound train has two gears fixed to the same shaft. Point to the connections to explain which arrangement you are looking at.

Read-only shaft-connection callout. An idler is a separate middle gear; a compound train has gears fixed together on one shaft. Do not infer compound merely from gear count.

Activity table: Drawing or response space

Accept an accurate labelled sketch of any described arrangement with a clear reason; no ratio required.

Student page 9

Complete the observations in words.

For a small driver and larger driven gear: slower output, opposite direction for one external pair. Same-size matching external gears: same speed, opposite directions. Large driver and smaller driven gear: faster output, opposite direction. Explanations should connect the size comparison to the visible movement, without ratios.

Compare a small driver turning a larger gear with a large driver turning a smaller gear. How is the output movement different?

Small driver to larger gear: slower output with more turning force. Large driver to smaller gear: faster output with less turning force.

Sketch one gear arrangement from the class example. Add labels and movement arrows, then explain its job in a sentence.

Look for driver, driven gear and shafts; arrows should reverse at each external mesh.

Activity table: Gear pair / Output: faster, slower or same? / Direction compared with driver / My explanation

For a small driver and larger driven gear: slower output, opposite direction for one external pair. Same-size matching external gears: same speed, opposite directions. Large driver and smaller driven gear: faster output, opposite direction. Explanations should connect the size comparison to the visible movement, without ratios.

Activity table: Drawing or response space

Small driver to larger gear: slower output with more turning force. Large driver to smaller gear: faster output with less turning force.

Activity table: Drawing or response space

Look for driver, driven gear and shafts; arrows should reverse at each external mesh.

Student page 10

How can a turning cam make another part move up and down?

The cam's changing shape pushes or lifts the follower as it turns. The follower has a return/contact arrangement.

Activity table: Drawing or response space

The cam's changing shape pushes or lifts the follower as it turns. The follower has a return/contact arrangement.

Student page 11

Activity table: Read movement and shape differently A movement line shows where the follower is during the cycle. It is not a cutting outline. Look at the physical cam and follower together. Your teacher's plan controls the actual shape, return arrangement and guards.

Read-only cam callout. A follower-position line is different from a physical cam outline. Use contact and return in the demonstration; no angle or graph calculations required.

Student page 12

Observe a cam in a teacher demonstration, drawing or video. Sketch the cam and follower and show where they touch. Add an arrow showing the cam turning and another showing the follower's movement. You may compare the two pictures if your teacher asks.

Observe a teacher-approved source. Sketch the cam and follower in contact, showing the turning cam and the follower's movement. Explain that shape changes move the follower. The actual return/contact arrangement must follow the demonstration.

Labelled cam sketch: turning part, follower, contact and movement arrows

Look for a recognisable turning cam, a labelled follower touching its profile, and clear arrows for turning and follower movement. Accept a labelled drawing, photo annotation or teacher-example observation; the picture is not a manufacturing template.

Explain what moves first and how that movement reaches the follower. What helps the follower return?

Cam turns first, shape moves follower, and the demonstrated spring or gravity arrangement keeps contact and returns it.

Activity table: Drawing or response space

Look for a recognisable turning cam, a labelled follower touching its profile, and clear arrows for turning and follower movement. Accept a labelled drawing, photo annotation or teacher-example observation; the picture is not a manufacturing template.

Activity table: Drawing or response space

Cam turns first, shape moves follower, and the demonstrated spring or gravity arrangement keeps contact and returns it.

Student page 13

What could happen if the pawl did not return into contact with the ratchet teeth?

The wheel could turn back because the catch may not engage.

Activity table: Drawing or response space

The wheel could turn back because the catch may not engage.

Student page 14

Use an approved picture or teacher demonstration, such as a ratchet socket tool. Draw the wheel and pawl. Add an arrow showing allowed movement. Mark where the pawl stops reverse movement. Explain what the catch does; you do not need to compare advanced bearing or clutch designs.

The student should use an identifiable teacher-approved real example. Look for accurate names and the one-way function. Do not require disassembly, operation of unfamiliar tools or a comparison with advanced bearings and clutches.

Real example: label the ratchet wheel, pawl and allowed direction

The wheel should have teeth and the pawl should be shown as a catch at the teeth. An arrow should identify allowed movement; a label should identify where a tooth blocks reverse movement. Accept a clear simplified diagram.

Explain how the wheel and pawl work together.

Pawl passes over teeth in the allowed direction and catches against a tooth when reverse movement is attempted.

Picture source or teacher demonstration, and one thing I learned:

Give the actual source title, usable link or a description identifying the teacher demonstration. The learning statement must be supported by the example, such as the pawl catching to prevent reverse movement.

Activity table: Drawing or response space

The wheel should have teeth and the pawl should be shown as a catch at the teeth. An arrow should identify allowed movement; a label should identify where a tooth blocks reverse movement. Accept a clear simplified diagram.

Activity table: Part / What it does / What I can point to

Pawl passes over teeth in the allowed direction and catches against a tooth when reverse movement is attempted.

Activity table: Drawing or response space

Give the actual source title, usable link or a description identifying the teacher demonstration. The learning statement must be supported by the example, such as the pawl catching to prevent reverse movement.

Student page 15

A motion-flow diagram uses arrows to show what drives what. Draw your actual proposed mechanism from handle to dispensing part. Label gears and shafts. Include a cam, gate or ratchet only where it belongs in the approved design. Describe what happens first, next and last.

Trace the actual approved arrangement from handle input to dispensing output. Include only the mechanisms present. Arrows and labels should show what drives what. Distinguish movement flow from an unrelated list of parts.

My mechanism: main parts, movement arrows and places fingers must stay clear of

Look for handle/input, connected gears or release parts, shafts and output; arrows should follow the demonstrated connections. Identify pinch-point areas or access routes that need teacher-approved protection. Do not invent components absent from the plan.

Describe the sequence for your proposed dispenser.

Describe a coherent first–next–last sequence using the actual design: ready, movement to release, release as intended, then ready again. A cam, gate or ratchet belongs in the explanation only when that design uses it.

What might happen if the gate opens before the lolly is in the right position?

The lolly may not release correctly or might become trapped; check the sequence in the demonstrated mechanism.

Activity table: Drawing or response space

Look for handle/input, connected gears or release parts, shafts and output; arrows should follow the demonstrated connections. Identify pinch-point areas or access routes that need teacher-approved protection. Do not invent components absent from the plan.

Activity table: Stage of the cycle / What should happen? / My explanation or check

Describe a coherent first–next–last sequence using the actual design: ready, movement to release, release as intended, then ready again. A cam, gate or ratchet belongs in the explanation only when that design uses it.

Activity table: Drawing or response space

The lolly may not release correctly or might become trapped; check the sequence in the demonstrated mechanism.

Student page 16

Describe one possible cause of a stiff handle. What safe step would you take next?

Possible rubbing, misalignment or overly tight gears; stop and ask teacher before making an approved adjustment.

Why should fingers stay clear of moving gears?

Gear teeth create pinch points and can trap fingers; keep clear and use required guards.

Activity table: Part or idea / Its job / What to check

Read-only function/check reference: handle provides input; gears and shafts pass movement; guard protects actual access points. It replaces the MA/VR/efficiency calculation table. No numerical answer required.

Activity table: Drawing or response space

Possible rubbing, misalignment or overly tight gears; stop and ask teacher before making an approved adjustment.

Activity table: Drawing or response space

Gear teeth create pinch points and can trap fingers; keep clear and use required guards.

Student page 17

1. What are the input and output of a hand-operated lolly dispenser?

Input: person turning handle. Output: movement of the dispensing part.

4. A small driver turns a larger driven gear. How does the output movement change?

Larger driven gear turns more slowly and has more turning force.

7. How can a cam and follower move a gate?

Turning cam lifts or pushes follower; follower operates gate; return/contact arrangement needed.

8. What does the pawl do in a ratchet mechanism?

Pawl allows the intended direction and blocks reverse movement. It does not control quantity by itself.

Activity table: Drawing or response space

Input: person turning handle. Output: movement of the dispensing part.

Activity table: Drawing or response space

Larger driven gear turns more slowly and has more turning force.

Activity table: Drawing or response space

Turning cam lifts or pushes follower; follower operates gate; return/contact arrangement needed.

Activity table: Drawing or response space

Pawl allows the intended direction and blocks reverse movement. It does not control quantity by itself.

Student page 18

List the main requirements and how you will check them.

Use the actual class brief to list function, what consistent use looks like, supplied materials/equipment, given dimensions or limits, and safety/user needs. Add an appropriate check for each. Do not invent numerical targets, trial counts or mandatory mechanisms.

Choose one requirement and explain a design decision that helps meet it.

Accept a reason linked to a real feature and a requirement from the approved brief.

Trace the movement from handle to output on your chosen sketch. Label the driver, driven gear and shafts, and add arrows showing direction. Use the gear arrangement and dimensions in the approved plan. You do not need to calculate ratios or design new tooth shapes. Show your teacher the idea before building.

A labelled sketch should identify driver, driven gear, shafts and movement direction, consistent with the approved arrangement. Actual dimensions must come from the plan. No tooth design, ratio or turning-force calculation is needed. Record genuine teacher feedback separately.

What will I change after feedback, and why?

Identify the actual feedback, the specific change proposed or made, and a reason. If feedback confirmed the design, explain the check that supported retaining it. Do not invent a teacher comment or claim approval not received.

Activity table: Requirement / constraint / Approved description or given size / How I will check

Use the actual class brief to list function, what consistent use looks like, supplied materials/equipment, given dimensions or limits, and safety/user needs. Add an appropriate check for each. Do not invent numerical targets, trial counts or mandatory mechanisms.

Activity table: Drawing or response space

Accept a reason linked to a real feature and a requirement from the approved brief.

Activity table: Drawing or response space

Identify the actual feedback, the specific change proposed or made, and a reason. If feedback confirmed the design, explain the check that supported retaining it. Do not invent a teacher comment or claim approval not received.

Student page 19

Sketch two different layouts before choosing. Show the handle, gears, container, outlet and base. Label movement with arrows and mark places fingers need protection. Include a cam or ratchet only if the approved design uses one. Compare ideas against the same class brief.

Two layouts should give meaningful choices, such as different handle/outlet positions or part placement within permitted design choices. Both should meet the same brief. A drawing need not be polished to communicate an accurate idea.

Idea A: label input, output and the path the lolly follows

Idea A should identify the input and output, the lolly path, main parts and movement. Show only teacher-approved mechanisms and dimensions. A labelled conceptual sketch is acceptable at this stage.

Idea B: a different layout to compare

Idea B should offer a real layout comparison, rather than simply recolouring Idea A. Labels and arrows should make the difference understandable. Stay within the actual brief and permitted design choices.

Which idea best meets the brief? Give a reason linked to a feature on your sketch.

Accept a justified choice using a feature such as stability, accessible handle, safe protection or build simplicity.

Activity table: Drawing or response space

Idea A should identify the input and output, the lolly path, main parts and movement. Show only teacher-approved mechanisms and dimensions. A labelled conceptual sketch is acceptable at this stage.

Activity table: Drawing or response space

Idea B should offer a real layout comparison, rather than simply recolouring Idea A. Labels and arrows should make the difference understandable. Stay within the actual brief and permitted design choices.

Activity table: Drawing or response space

Accept a justified choice using a feature such as stability, accessible handle, safe protection or build simplicity.

Student page 20

Compare the supplied material options for useful parts.

Compare the materials actually supplied or considered with the teacher. Name a useful property, a suitable part and a reason. For example, an approved plywood base may provide stable support; approved clear acrylic may allow visibility. Material choice alone does not establish food-contact suitability.

Complete your approved parts list.

List actual parts, quantities, approved materials and given sizes. Check the list against the drawing and keep a usable complete-list reference if needed. No cost or gear-geometry calculation is required. Do not invent missing stock or dimensions.

Activity table: Approved material / A useful property / A suitable part / Reason for my choice

Compare the materials actually supplied or considered with the teacher. Name a useful property, a suitable part and a reason. For example, an approved plywood base may provide stable support; approved clear acrylic may allow visibility. Material choice alone does not establish food-contact suitability.

Activity table: Part and quantity / Approved material / Size from the plan / Drawing or list reference

List actual parts, quantities, approved materials and given sizes. Check the list against the drawing and keep a usable complete-list reference if needed. No cost or gear-geometry calculation is required. Do not invent missing stock or dimensions.

Student page 21

Sheet plan: label the parts, useful offcuts and grain direction where it matters

Show the actual usable sheet or an explicitly illustrative class layout. Label intended parts, useful offcuts and grain direction where relevant. Follow teacher spacing and process instructions. Avoid damaged stock and maintain safe support; maximum packing is not the only goal.

Explain two choices that reduce waste while keeping the parts suitable for their jobs.

Suitable part arrangement, useful offcut reuse, avoiding damaged material and following grain/spacing instructions.

My complete parts-list file or page reference:

Provide a usable file name or page reference for the actual complete parts list, with revision information if used. It should allow the teacher or student to locate the correct list. No invented price source is required.

Activity table: Drawing or response space

Show the actual usable sheet or an explicitly illustrative class layout. Label intended parts, useful offcuts and grain direction where relevant. Follow teacher spacing and process instructions. Avoid damaged stock and maintain safe support; maximum packing is not the only goal.

Activity table: Drawing or response space

Suitable part arrangement, useful offcut reuse, avoiding damaged material and following grain/spacing instructions.

Activity table: Drawing or response space

Provide a usable file name or page reference for the actual complete parts list, with revision information if used. It should allow the teacher or student to locate the correct list. No invented price source is required.

Student page 22

Drawing or CAD screenshot: label the main components and movement

A hand drawing or basic CAD screenshot should label the main components and show input/output movement with arrows. CAD is optional when the teacher directs another method. Accuracy and clarity matter more than advanced features.

Detail sketch or screenshot: show how two parts fit together

Show a meaningful joining or fitting detail from the actual design. Label the two parts and explain which is fixed or moves. A careful sketch is a valid alternative to CAD. The detail should be consistent with the approved plan.

Identify the main parts and their jobs.

Identify the main part groups actually used, their jobs and a suitable check. Typical groups are gears, the release part and the frame/base/guard. Do not require a cam or ratchet absent from the actual plan.

Drawing or CAD file name, and one thing I checked:

Give a usable actual drawing or CAD file name. State one real check, observation or question, such as whether parts line up or have room to move. If not yet checked, identify the next check honestly.

Activity table: Drawing or response space

A hand drawing or basic CAD screenshot should label the main components and show input/output movement with arrows. CAD is optional when the teacher directs another method. Accuracy and clarity matter more than advanced features.

Activity table: Drawing or response space

Show a meaningful joining or fitting detail from the actual design. Label the two parts and explain which is fixed or moves. A careful sketch is a valid alternative to CAD. The detail should be consistent with the approved plan.

Activity table: Main part or group / Its job / What I will check

Identify the main part groups actually used, their jobs and a suitable check. Typical groups are gears, the release part and the frame/base/guard. Do not require a cam or ratchet absent from the actual plan.

Activity table: Drawing or response space

Give a usable actual drawing or CAD file name. State one real check, observation or question, such as whether parts line up or have room to move. If not yet checked, identify the next check honestly.

Student page 23

Label a drawing or screenshot with movement arrows. If using CAD, follow the teacher's demonstration to check that fixed parts stay still and moving parts can move as intended. A screen recording or advanced motion study is not required.

Use a labelled drawing, CAD screenshot or clearly identified demonstration. Arrows should show movement; fixed and moving parts should be distinguished. No recording, motion-study tool or advanced simulation is required. A digital view does not authorise practical use.

Record what the picture or demonstration helps you understand.

State what the image or demonstration actually supports, such as where the handle movement goes or where a part may collide. Separate a visible observation from a possible cause. Do not claim a physical test from a static picture.

A model shows one part overlapping another. What would you check before making the parts?

Check the approved drawing, positions and room to move, then consult teacher before making or changing parts.

Activity table: Drawing or response space

Use a labelled drawing, CAD screenshot or clearly identified demonstration. Arrows should show movement; fixed and moving parts should be distinguished. No recording, motion-study tool or advanced simulation is required. A digital view does not authorise practical use.

Activity table: What to look at / What I notice or understand

State what the image or demonstration actually supports, such as where the handle movement goes or where a part may collide. Separate a visible observation from a possible cause. Do not claim a physical test from a static picture.

Activity table: Drawing or response space

Check the approved drawing, positions and room to move, then consult teacher before making or changing parts.

Student page 24

Write the main stages of your build plan.

List main stages in the teacher's construction sequence: check instructions/safety, prepare parts, check fit, assemble, check movement/protection, teacher inspection before testing. Exact local order takes precedence over a generic example.

Choose one possible problem and a change to discuss with the teacher.

Name a plausible problem, a specific change to discuss, and why it could help. For example, a rubbing gear may need its position checked; the student should not invent a fit allowance or modify it without teacher guidance. Accept other evidence-based problems.

Why is it useful to check the fit before final assembly?

It catches rubbing or a loose joint before parts become difficult to adjust after assembly.

Activity table: Build stage / What I will do / Check before moving on

List main stages in the teacher's construction sequence: check instructions/safety, prepare parts, check fit, assemble, check movement/protection, teacher inspection before testing. Exact local order takes precedence over a generic example.

Activity table: Problem I might find / Change to discuss with the teacher / Why it might help

Name a plausible problem, a specific change to discuss, and why it could help. For example, a rubbing gear may need its position checked; the student should not invent a fit allowance or modify it without teacher guidance. Accept other evidence-based problems.

Activity table: Drawing or response space

It catches rubbing or a loose joint before parts become difficult to adjust after assembly.

Student page 25

Plan a useful front or top view: labels, movement arrows and given dimensions

A front or top view should show the useful arrangement, readable part/material labels, movement arrows and the actual given dimensions needed to build. Additional views are useful only when they clarify hidden information. No numerical gear design is required.

Record and review your final drawing.

Provide a usable drawing reference, explain which views/labels/dimensions were checked, and record real feedback or a next action. The drawing can be hand drawn or basic CAD as directed. Do not invent a completed review.

What did another reader find unclear, and how will you improve it?

Accept actual reader feedback and a specific correction such as a clearer label or view.

Activity table: Drawing or response space

A front or top view should show the useful arrangement, readable part/material labels, movement arrows and the actual given dimensions needed to build. Additional views are useful only when they clarify hidden information. No numerical gear design is required.

Activity table: Drawing or file name / Views, labels and given dimensions / Feedback and next step

Provide a usable drawing reference, explain which views/labels/dimensions were checked, and record real feedback or a next action. The drawing can be hand drawn or basic CAD as directed. Do not invent a completed review.

Activity table: Drawing or response space

Accept actual reader feedback and a specific correction such as a clearer label or view.

Student page 26

Check that your design pack is easy to follow.

The pack should contain the brief/checklist, labelled mechanism sketch, materials/parts list, sheet plan where used, final drawing and build plan. CAD screenshots or manufacturing files are conditional on actual class use. Check that references are usable and versions agree.

If another person used your design pack, what would be easy to understand and what might need a clearer label?

Accept a realistic judgement about the clarity of the brief, sketch, dimensions, materials or build sequence.

Activity table: Design-pack item / File or page / What I checked

The pack should contain the brief/checklist, labelled mechanism sketch, materials/parts list, sheet plan where used, final drawing and build plan. CAD screenshots or manufacturing files are conditional on actual class use. Check that references are usable and versions agree.

Activity table: Drawing or response space

Accept a realistic judgement about the clarity of the brief, sketch, dimensions, materials or build sequence.

Student page 27

7. What information should a final design drawing include?

Clear views, part labels, materials and approved dimensions; movement arrows and joining notes where helpful.

8. What is the difference between a CAD part and a CAD assembly?

A part is one component. An assembly shows how components fit together.

9. Why should your build plan put steps in a sensible order?

A sensible sequence allows access to fit parts and places checks before final assembly, reducing rework.

Activity table: Drawing or response space

Clear views, part labels, materials and approved dimensions; movement arrows and joining notes where helpful.

Activity table: Drawing or response space

A part is one component. An assembly shows how components fit together.

Activity table: Drawing or response space

A sensible sequence allows access to fit parts and places checks before final assembly, reducing rework.

Student page 28

Identify hazards and the safety checks your teacher requires.

Identify hazards from the actual approved operation, their possible consequences and the controls the teacher requires. Suitable areas to consider include cutting, drilling, dust, sharp edges, moving parts and clearing a jam. The record must reflect the class procedure; a generic answer does not authorise equipment use.

Check the drawing and file before making a part.

Check that the current drawing and manufacturing file show the same intended version, correct units and size, approved material and required teacher checks. Where computer cutting is not used, record the drawing and manual-process checks instead.

Where will you find your class workshop instructions? When would you stop and ask for help?

Name the actual teacher-issued workshop instructions or demonstrated procedure and where they are kept. Stop for unexpected binding, breakage, damage, movement or another unsafe condition; keep clear and seek teacher help. The local stop method must come from the teacher.

Activity table: Planned operation / Hazard / consequence / Teacher-approved control / check

For each operation used, connect a real hazard to its consequence and the specific teacher-required control. Controls and permission must come from the actual class process. Do not invent machinery settings or practical sign-off.

Activity table: Check with the teacher / Why it matters / My evidence / status

Record current file/version agreement, checked size/units, teacher review of cutting file, correct material and approved setup/holding where applicable. Use actual status or an honest next step rather than a blanket “safe”.

Student page 29

Record a teacher-directed fit check.

Record the actual part or scrap trial, what fit was intended, what was observed and the teacher-guided next step. A successful fit or a fit problem is acceptable if honestly described. If not performed, state that rather than inventing a result.

Why is it useful to check a fit before making or assembling every part?

A fit check can reveal a tight, loose or misaligned part while it is easier to correct. It can prevent repeating the same mistake across several parts and reduce wasted material.

Activity table: Part or scrap test / What I noticed / Teacher-guided next step

Identify the actual scrap or part, intended fit, observed fit and teacher-guided change or recheck. It is acceptable to record “not yet performed”. No kerf arithmetic is expected.

Student page 30

Label a sketch or drawing: reference edge, shaft positions and checks.

Label a real approved drawing or sketch with the edge used for marking out, intended shaft positions and the checks to make. Labels should connect to the actual class plan rather than supply guessed dimensions.

Record the hole-position and alignment checks you made.

Show where the hole positions were checked against the approved drawing and whether supports or shaft positions lined up. Record the actual observation and any teacher-guided action; do not invent a numerical tolerance.

What might happen if a shaft hole is in the wrong place? Explain the effect on the moving parts.

A misplaced hole can make a shaft lean or put gears in the wrong position. The gears may rub, catch or fail to meet properly, so the handle may become stiff. Accept another clear cause-and-effect chain supported by the mechanism.

Add a progress-photo reference and the checks you made.

Give a usable photo or file reference and a short caption explaining the work or check shown. The photo must be the student's actual progress or clearly identified teacher demonstration, not an unlabelled illustration.

Activity table: Feature to check / Drawing or teacher guidance / What I noticed / next step

Locate shafts and holes on the approved drawing; record whether positions line up, the hole/shaft fit and matching orientation. Actual dimensions come from the approved plan, not the answer guide.

Student page 31

Record what you prepared, assembled and checked.

Describe the actual surface preparation, assembly step and checks made. Look for named parts, a clear observation and any next action. Approved preparation, correct alignment and free movement are relevant concepts; the actual method remains teacher-controlled.

What would you look for to check that the gears move freely?

With the teacher's safe checking method, look for gears meeting properly, shafts staying in position and no gear rubbing against the frame or another part. Movement should be smooth rather than forced. The approved plan controls actual fit and clearance.

Activity table: Part or feature / Observation or photo / Next step and recheck

For edges, frame/shafts, gears and complete assembly, record an actual observation/photo and a next step. Evidence should explain function: smoother surfaces reduce catching; alignment helps movement; retention keeps parts in position.

Student page 32

Label a photo or sketch: moving parts, finger-access points and guards.

Identify actual moving parts and pinch points, show where fingers could approach them, and label the guard or other teacher-approved protection. A transparent cover alone is not proof that every access path is protected.

Record the movement, guard and pre-test checks.

Record the actual gate movement and return, any direction control used, guard coverage, secure fixings, edges and stable base. State any concern and the teacher-approved next step. Teacher sign-off is a separate teacher action, not something a written answer can claim.

What will you do if the dispenser jams?

Stop, keep fingers clear and ask the teacher for help. Do not force the handle or reach into the mechanism. Follow the teacher's stop and safe fault-checking procedure.

Activity table: Check / What I noticed / Teacher-approved next step

Record the actual cam/gate operation and return, ratchet direction control where used, guard coverage, secure fixings, edges and stable base. Identify a concern and stop for teacher help where needed. Student writing cannot substitute for teacher safety approval.

Student page 33

Record your building progress and changes.

Use dated entries saying what was made, what was checked, a problem or change and the next step. Add useful photo references and explain changes from the drawing. Accept an entry with no problem if it records a meaningful check honestly.

Show your build record to the teacher before testing.

Locate the actual build record and the feedback received before testing. Record an unresolved issue if one remains. The student must not invent teacher approval.

Activity table: Date / part or stage / What I made or checked / Photo or file / Problem, change or next step

Dated entries should identify what was made or checked, a useful photo/file, the problem or change and a next step. A meaningful successful check is valid; a fault need not be invented.

Activity table: Item / Where it is / next step

Provide usable references to build log/photos and current drawing/build records, plus actual movement, edge and guarding checks and teacher feedback. Mark missing checks honestly.

Student page 34

1. Why should you use the approved drawing when making a part?

It gives the intended shape, size and position of features such as holes. Guessing from a photo can leave parts that do not fit together.

4. A gear rubs against the frame. What should you do?

Stop and show the teacher where it rubs. Check its position and the space around it as directed. Do not force the handle.

10. What should a useful build-log entry include?

Say what you made, what you checked, any problem or change, and your next step. A labelled photo can help explain your work.

Student page 35

Plan what you will observe and how you will test safely.

Identify what normal operation should look like, the teacher-approved pieces and method, observations to make, how the comparison will be kept fair, and when to stop. All safety and participant decisions must reflect teacher instruction.

Explain what a normal release looks like and how you will recognise a jam or an incomplete release.

Describe the agreed normal release in the actual class design. A jam means an item or part catches so intended movement or release is blocked; an incomplete release means the intended job has not finished. Distinguish these using observations rather than a success percentage.

Activity table: Test question / My teacher-approved plan

Define intended normal operation, teacher-approved pieces, relevant observations, a fair repeated method and the local stop/help procedure. No fixed trial count or percentage target is required.

Student page 36

Record what you notice and the next safe step.

Record what the handle, gears, gate and base actually do. Useful wording identifies smooth movement, a catch, rubbing or wobbling and its location. The next step should be safe, such as asking the teacher to inspect a fault. Do not reward invented faults or unperformed tests.

Activity table: Part or feature / What I noticed / Next safe step

Record qualitative observations for handle, gears/shafts, gate/feed, frame/base, guards/edges and another relevant feature. Identify location and behaviour; state the next safe step. No observations may be fabricated to fill rows.

Student page 37

Record one change and your before-and-after observations.

Name one teacher-approved adjustment, describe the original problem and compare the actual observations before and after using the same pieces and method. An unchanged or worse result is valid evidence if honestly recorded.

Activity table: What I checked / Before the change / After the change

Name one approved change and compare actual before-and-after behaviour using the same pieces and method. Explain handle motion, release, gate return, stability/ease where relevant. An unchanged or worse result remains valid evidence.

Student page 38

Summarise your observations in words.

Summarise actual movement and dispensing observations in clear words. Explain what happened and where. Reliability is consistency during repeated use; no numerical rate or calculation is expected.

Compare what you expected with what you observed.

State a prediction in words and compare it with an actual observation for the same action or part. Explain an agreement or difference. Separate what was observed from a suggested cause that still needs checking.

What does your observation show? What would you need to check again?

Explain what the recorded observation supports, then identify a sensible follow-up. For example, a smooth test supports smooth movement in those conditions but does not prove it will never jam. Any example must be labelled as illustrative rather than the student's result.

The handle turns but a lolly does not come out. Suggest a possible cause and a safe check you could discuss with the teacher.

Possible causes include a lolly catching before the outlet, a gate not opening correctly, or the gears moving without the intended gate movement. Discuss a stopped, teacher-directed visual check of the relevant part. Do not reach into a moving mechanism or force it. Accept another plausible cause and safe check.

Activity table: What I tested / Movement / Release / Problem / Next step

Use words for movement, release and problems and identify a safe next step. The summary should match actual observations, not be turned into totals or a reliability calculation.

Activity table: Part or action / What I expected / What I observed

For handle, gears, gate and release, make a prediction in words and compare it with a real observation. A clear discrepancy may suggest a next check but is not proof of a cause.

Student page 39

My labelled sketch or photo: the problem, its location and a possible change

Use an actual photo or a clearly labelled sketch. Point to the location of an observed problem, describe what happened and distinguish an improvement proposal from a completed change. Do not require invented damage or a numeric dimension.

Explain what your image shows and why the proposed change might help.

Explain what is visible and connect the proposed change to the observed problem. A good response says why the change might help without claiming it has already worked. Actual geometry and method require teacher approval.

Student page 40

Describe a real problem or a feature you checked.

Name a real observed problem and describe its location and behaviour. If no problem was found, state a feature checked and the actual observation. A possible cause is a hypothesis until checked.

Explain a change you tried or would discuss with the teacher.

Explain one teacher-approved change actually tried, or clearly label a proposed change for discussion. Link it to the observed problem and explain the intended effect. Do not imply that a proposal was carried out or approved.

Suggest a realistic next improvement and explain why it could help.

Suggest a practical improvement linked to an observed need and explain why it might help. For example, a direction arrow could help a user who was unsure which way to turn. Accept different feasible ideas; teacher approval controls physical alterations.

Activity table: What happened? / Possible cause and observation / One safe check or change

Separate “what happened” from “possible cause”. Suggest a safe, teacher-guided check or change connected to the observation. Accept multiple plausible causes if justified.

Activity table: My improvement / Change I tried / Another idea, if useful

Link the actual problem to one approved change, explain why it might help, locate evidence of the change, compare the same test before/after and describe the result honestly. A proposed change must be identified as not yet tested.

Activity table: Improvement idea / Problem it could solve and why

A useful improvement solves an observed problem, is practical in the class context and has a clear explanation. Extra ideas are optional; quantity alone does not demonstrate quality.

Student page 41

Add or plan a labelled photo of the finished dispenser.

Show the actual completed dispenser or plan where the student will place its photo. Labels should identify the handle, gears, gate and guard where those exist in that design. Do not require a component absent from the approved project.

Link the change in your image to what you observed in testing.

Identify the visible changed feature and connect it to the before-and-after test observation. A photo shows appearance or position, while the recorded observation explains the effect during use. Do not treat an image alone as proof of performance.

Write a caption explaining one feature and how it helps the dispenser work.

A useful caption names the visible feature and explains its job, such as a guard helping keep fingers from a moving gear. Accept another accurate explanation of an actual visible feature.

Activity table: Photo or sketch / Label what changed / What I noticed in testing

Locate comparable before/after photos or sketches, label what changed and link to a real test observation. The images must not be presented as proof of an unobserved performance claim.

Student page 42

Record useful comments from an approved user trial.

Record a genuine comment from a teacher-approved participant about comfort, clear direction, easy collection or another relevant user experience. Use simple participant labels and avoid unnecessary personal information. An unperformed trial must be recorded as such.

What did the user notice? Explain one change their comment suggests.

Quote or accurately paraphrase an actual user comment and suggest a related improvement or next check. Explain the connection. A single user's comment is useful feedback, not proof that all users will have the same experience.

If a user is unsure which way to turn the handle, how could the design or instructions help?

A clear direction arrow or short operating instruction could help show the intended direction. It should match the actual mechanism and teacher instructions. Do not assume a ratchet replaces clear communication or that the user should be forced to discover the direction.

Activity table: User / Comfort and ease / What was clear or confusing / Suggested change

Record actual approved participant comments about comfort/ease, clarity and possible improvements. A single user is enough for the revised learning; additional participants are optional and teacher-controlled.

Student page 43

Use actual observations to judge the design.

Judge the design against the brief using observations, labelled photos or user comments. Explain a strength and an area to improve. Use the table to consider function, safety features, ease of use, build quality, appearance and careful material use. Safety conclusions must remain within teacher checks.

Write a short evaluation: "My dispenser … because I observed … . I would improve … because … ."

Include a judgement, an actual observation supporting it and a realistic improvement with a reason. Illustrative wording only: "The dispenser was easy to use because the handle moved smoothly during our approved test. I would add a direction arrow because the user was unsure which way to turn." Students must substitute their own evidence.

Activity table: Design question / What I observed / My judgement / improvement

Use actual observations to judge function, safety features, ease of use, build quality, appearance and careful material use. A strength and a realistic improvement must be explained. Avoid asserting universal safety or durability from a short classroom test.

Student page 44

Explain how your project used materials carefully.

Describe an actual material-use choice, such as careful cut planning, reusing a suitable offcut with approval, avoiding a remake or designing a replaceable part. Explain how it reduced waste or could extend useful life. Do not claim verified environmental benefits without evidence.

Suggest a way to reduce waste or make a part easier to repair. Explain why it could help.

Suggest a feasible lower-waste or repair idea and explain the reason. For example, checking marking out before cutting may avoid remaking a part. The idea must preserve strength, fit and the teacher-approved safe process.

Activity table: Project choice / What I noticed / Possible improvement

Evaluate actual cut/offcut planning, remade parts, approved finish/use and repairability. Explain an effect or uncertainty and a realistic improvement; do not assume "timber = sustainable".

Student page 45

Plan a short showcase explanation or display.

Plan a clear message about purpose, mechanism, an actual test observation, a safety feature and one improvement. Use actual final photos and any demonstration method approved by the teacher. The showcase does not replace safety approval.

Write a short explanation a visitor can understand. Include what worked, how you know and one improvement.

Explain the dispenser to a visitor in ordinary language. Identify what it does, how a relevant mechanism helps, what actually worked or needs improvement, and the observation behind the judgement. No numerical result is required.

Activity table: Showcase idea / What I will show or say

Plan how to communicate purpose, mechanism operation, an actual test observation, an improvement and safety/material-use evidence. Ordinary language and labelled visuals are appropriate.

Student page 46

“I understand that … . A challenge I worked through was … . I did this by … .”

Identify a mechanism concept now understood and a real challenge. Explain what action, feedback or check helped. Accept individual experiences, including an unresolved difficulty described honestly; do not provide invented personal evidence.

“A skill I improved was … . Next time I would … because … .”

Name a skill that improved, give a real example and explain a useful next-time change. Link the change to what the student learned rather than a generic claim that they would “do better”.

Student page 47

List where each part of your folio can be found.

Provide usable page numbers or file references for each required folio section. Locations should lead to the actual current records. The index should include design, build, testing, evaluation, improvement and reflection evidence requested by the teacher.

What feedback did you use? Is there a question you still need to ask the teacher?

Identify actual feedback, the action taken and any remaining question. State when feedback has not yet been received. Do not invent teacher comments or approval.

Current teacher instructions for submission:

Record the current teacher-issued format, destination and timing. The teacher's actual instruction controls this entry; the workbook does not establish a deadline or submission policy.

Activity table: Folio section / Page or file / Ready / next step

Supply usable current page/file references for cover/brief/criteria, mechanism explanations, materials/drawings, safety/build records, observations/feedback, evaluation/reflection and showcase/submission instructions. Mark unfinished items accurately.

Student page 48

1. What does reliability mean for your dispenser?

It means how consistently the dispenser does its job during repeated use. Describe whether it keeps working smoothly or has recurring problems.

8. How is evaluation different from describing your build?

Description says what you made or did. Evaluation judges how well it worked and explains why, using observations or user feedback.

9. Improve this sentence: “My dispenser is good.”

For example: “My dispenser was easy to use because the handle turned smoothly and my classmate could collect the lolly easily.” The observation explains the judgement.

10. What makes an improvement useful?

It solves a problem you observed and could realistically be made. For example, a direction arrow could help a user who was unsure which way to turn the handle.

Student page 49

Continue questions 2, 3, 5, 6, 7, 8 and 9.

Complete current Toolkit 3 questions 2, 3, 5, 6, 7, 8 and 9. Printed questions are 1, 4 and 10, so these sets cover all ten without repetition or omission.

Continue questions 2, 3, 4, 5, 6 and 7.

Complete current Toolkit 4 questions 2, 3, 4, 5, 6 and 7. Printed questions are 1, 8, 9 and 10, so these sets cover all ten without repetition or omission.

Teacher-named place to save your responses:

Name the exact teacher-requested place where the responses are saved. This is a file-location record, not evidence that a task was formally submitted.

Student page 50

Teacher-led demonstration: title, something I noticed and a question about my own design.

Identify the actual demonstration or source, record something genuinely noticed and ask a relevant question about the student's own design. Demonstration settings or dimensions do not automatically apply to the class project.

Activity table: Term / Meaning in this project

Use the simplified glossary in explanations: driver starts movement; driven receives it; gear size can change speed; friction resists rubbing; alignment means parts line up; a cam pushes a follower; clearance provides movement space; reliability means consistency; a reference edge supports marking out and the current version identifies the agreed file.

Website concept checks

The following 40 model responses match the live toolkits. Fourteen of these checks appear in the printed workbook; the continuation page links to the rest.

Toolkit 1

1. What are the input and output of a hand-operated lolly dispenser?

The input is the person turning the handle. The output is the movement of the dispensing part, which helps release the lolly.

2. What is the difference between a driver gear and a driven gear?

The driver provides the turning movement. Its teeth push the teeth of the driven gear, which receives that movement.

3. If two external gears mesh and the driver turns clockwise, which way does the driven gear turn?

It turns anticlockwise. Two meshing external gears turn in opposite directions.

4. A small driver turns a larger driven gear. How does the output movement change?

The larger driven gear turns more slowly and has more turning force. We can describe the change without calculating it.

5. What does an idler gear do in a simple three-gear line?

It passes movement between the first and last gears. Each meshing pair reverses direction, so the first and last gears turn in the same direction.

6. What makes a gear train a compound gear train?

It has two gears fixed to the same shaft so they turn together, linking more than one gear pair. Simply adding another separate gear does not make it compound.

7. How can a cam and follower move a gate?

As the cam turns, its changing shape pushes or lifts the follower. The follower moves the gate. The demonstrated design needs a way to keep contact and return the follower.

8. What does the pawl do in a ratchet mechanism?

It is a catch that passes over teeth in the allowed direction and blocks reverse movement. Other parts control how much is dispensed.

9. Why might a gear system be difficult to turn?

The gears may be too tight, a shaft may be crooked, or a part may be rubbing. Stop and ask the teacher before adjusting it; forcing the handle can damage the parts.

10. Why should fingers stay clear of moving gears?

The teeth can draw fingers into a pinch point. Keep clear, use the required guards and follow the teacher's safety instructions.

Toolkit 2

1. What is a design brief, and how does it help you start?

It explains the problem and the project requirements. Reading it helps me plan something that does the required job using the approved materials and parts.

2. Why is it useful to sketch more than one idea?

It gives me options to compare. I can choose a layout that meets the brief and explain why it will be easier to make or use.

3. What should movement arrows on your sketch show?

They should show which way turning parts rotate or which way a sliding part moves. They help another person follow the movement from input to output.

4. What is clearance, and why does a moving gear need it?

Clearance is the space needed to move without unwanted rubbing or collision. A gear needs room to turn while its teeth still mesh correctly with the next gear.

5. Why should you check the alignment of shafts and gears?

Correct positions help the gears mesh and turn smoothly. Crooked shafts or misaligned gears can cause rubbing, slipping or a jam.

6. How would you justify a material choice for the base?

I would name the approved material and link a useful property to the job. For example, the supplied plywood can provide a stable support for the project.

7. What information should a final design drawing include?

It should show clear views, part labels, materials and the dimensions needed to build from the plan. Movement arrows and joining notes make it easier to follow.

8. What is the difference between a CAD part and a CAD assembly?

A part is one component, such as a gear or base. An assembly shows how components fit together.

9. Why should your build plan put steps in a sensible order?

Some parts are easier to fit before other parts enclose them. A clear order helps avoid redoing work and reminds me when to check fit, safety and movement.

10. What should you do before changing a dimension or part on the supplied plan?

Check the reason for the change with my teacher. One change can affect how other parts fit or move, so I should not guess.

Toolkit 3

1. Why should you use the approved drawing when making a part?

It gives the intended shape, size and position of features such as holes. Guessing from a photo can leave parts that do not fit together.

2. What is the difference between CAD and CAM?

CAD helps you draw or model the design on a computer. CAM uses computer information to help a machine make a part.

3. Why is a trial fit useful before final assembly?

It lets you find a part that does not fit or line up while it is still easier to adjust. Follow the assembly process shown by your teacher.

4. A gear rubs against the frame. What should you do?

Stop and show the teacher where it rubs. Check its position and the space around it as directed. Do not force the handle.

5. What does alignment mean?

It means parts are lined up in the right positions. For example, lined-up shafts help the gears meet and turn properly.

6. Why does a moving part need clearance?

It needs space to move without rubbing or catching on another part. The required fit comes from your approved plan and teacher guidance.

7. How can a rough edge affect the mechanism?

It may rub or catch on another part, adding friction and making movement stiff. Finish edges using the method demonstrated in class.

8. What is the purpose of a guard?

A guard helps keep fingers away from moving parts and pinch points. It must be fitted and checked as directed before use.

9. Why should you label your parts?

Labels help you recognise each part and its position. This makes assembly easier and helps avoid fitting a part the wrong way around.

10. What should a useful build-log entry include?

Say what you made, what you checked, any problem or change, and your next step. A labelled photo can help explain your work.

Toolkit 4

1. What does reliability mean for your dispenser?

It means how consistently the dispenser does its job during repeated use. Describe whether it keeps working smoothly or has recurring problems.

2. Why should you repeat a test?

A single successful try may miss a problem. Repeating the same test helps you notice whether it works consistently and whether a fault happens again.

3. What should you do if the mechanism jams?

Stop, keep fingers clear and ask the teacher for help. Do not force the handle or reach into moving parts.

4. What is a useful observation to record?

For example: “The gate caught as it closed” or “The handle turned smoothly.” A clear observation describes what happened and where.

5. Why should you change one thing at a time?

It helps you tell which change affected the result. If you change several things together, you cannot be sure which one helped.

6. How can you make a before-and-after comparison fair?

Use the same lollies and the same testing method. Change only the part you are investigating, with teacher approval.

7. Why ask someone else to try your dispenser?

Another user may notice an uncomfortable handle, confusing direction or awkward outlet that you missed. Their comments help you judge ease of use.

8. How is evaluation different from describing your build?

Description says what you made or did. Evaluation judges how well it worked and explains why, using observations or user feedback.

9. Improve this sentence: “My dispenser is good.”

For example: “My dispenser was easy to use because the handle turned smoothly and my classmate could collect the lolly easily.” The observation explains the judgement.

10. What makes an improvement useful?

It solves a problem you observed and could realistically be made. For example, a direction arrow could help a user who was unsure which way to turn the handle.